Term Work Module: **15 HOURS**

Objective: The Term Work Module aims to provide practical understanding of software testing concepts, testing documentation, and manual testing techniques through hands-on activities and real-world testing scenarios. It also helps students analyze different testing approaches and develop Selenium automation scripts using Python for web application testing, validation, and regression testing.

Term Work	Statement	Aligned Practical/Tutorial No.	Reason for Alignment
Term Work 1 (3 Hours)	Understand SDLC and STLC phases, testing activities, requirement reviews, and prepare Test Plan, RTM, and Testing Strategy.	**Tutorial 1, Tutorial 2, Tutorial 3**	Covers SDLC/STLC, testing activities, Test Plan, RTM, Testing Strategy, requirement reviews, walkthroughs, inspections, and peer reviews.
Term Work 2 (2 Hours)	Apply functional, non-functional, positive, and negative test cases and prepare professional bug reports.	**Tutorial 4, Tutorial 5**	Covers functional/non-functional testing, positive & negative test cases, defect identification, and bug report preparation.
Term Work 3 (3 Hours)	Apply BVA, Equivalence Partitioning, manual test execution, and defect management using Bugzilla and TestLink.	**Tutorial 6, Tutorial 7, Tutorial 8**	Includes manual test execution, execution reports, BVA, Equivalence Partitioning, Bugzilla, TestLink, and defect management.
Term Work 4 (2 Hours)	Analyze black-box, white-box, integration, system testing, and AI-powered software testing tools.	**Tutorial 9, Tutorial 10, Tutorial 11, Tutorial 12**	Covers testing levels, automation suitability analysis, Python automation concepts, regression testing, and AI-powered software testing tools.

Term Work 5 (5 Hours)	Create Selenium testing environment and develop Selenium automation scripts for browser navigation, login authentication, form handling, assertions, validation, and exception handling.	**Tutorial 13, Tutorial 14, Tutorial 15**	Covers Selenium environment setup, Python configuration, browser drivers, Selenium scripting, login automation, assertions, validation, and exception handling.

Term Work	Tutorial(s)	Student Tasks	Deliverables	Justification
Term Work 1 (3 Hours)	**Tutorial 1, 2, 3**	• Select a sample software project (Library, Hospital, Banking, E-Commerce, etc.) • Identify SDLC & STLC phases • Map testing activities to each SDLC phase • Study Software Requirement Specification (SRS) • Perform requirement review and peer review • Prepare Test Plan • Prepare Requirement Traceability Matrix (RTM) • Prepare Testing Strategy document	• SDLC-STLC Mapping Report • Requirement Review Checklist • Test Plan • RTM • Testing Strategy	Students first understand the complete testing lifecycle before writing or executing test cases. These documents are standard Software Testing artifacts used in industry.
Term Work 2 (2 Hours)	**Tutorial 4, 5**	• Identify functional and non-functional requirements • Design positive and negative test cases for Login & Registration modules • Execute test cases manually • Identify defects	• Functional Test Cases • Non-functional Test Cases • Positive & Negative Test Cases • Bug Report Sheet	After understanding documentation, students learn how software is validated and how defects are reported using industry practices.

		• Capture screenshots • Assign Severity & Priority • Prepare professional Bug Reports		
Term Work 3 (3 Hours)	**Tutorial 6, 7, 8**	• Execute manual test cases • Record Pass/Fail status • Prepare Test Execution Report • Prepare Test Summary Report • Apply Boundary Value Analysis (BVA) • Apply Equivalence Partitioning (EP) • Create and manage defects using Bugzilla • Manage test cases using TestLink	• Manual Test Execution Report • Test Summary Report • BVA Table • EP Table • Bugzilla Screenshots • TestLink Screenshots	Students move from designing test cases to executing them using professional defect and test management tools, simulating an industrial QA process.
Term Work 4 (2 Hours)	**Tutorial 9, 10, 11, 12**	• Perform Black-box, White-box, Integration, and System Testing analysis • Identify modules suitable for automation testing • Analyze regression testing scenarios • Explore AI-powered testing tools • Generate AI-based test cases • Generate AI-based test data • Perform defect prediction • Generate Selenium scripts using Generative AI	• Testing Technique Comparison Report • Automation Feasibility Report • AI Testing Tool Analysis Report • AI-generated Test Cases & Test Scripts	Students explore advanced testing techniques and modern AI-assisted software testing practices that are increasingly adopted in industry.

Term Work 5 (5 Hours)	**Tutorial 13, 14, 15**	• Install Python, Selenium, VS Code/PyCharm, Browser Drivers • Configure Selenium environment • Automate browser navigation • Automate Login page • Automate Registration Form • Locate web elements using different locators • Perform Assertions • Validate outputs • Handle exceptions • Generate execution report	• Environment Setup Screenshots • Selenium Python Programs • Execution Output • Validation Report • Automation Test Report	This term work enables students to develop complete Selenium automation scripts, progressing from environment setup to real-world web application testing using Python.

Overall Progression

Term Work	Learning Focus
Term Work 1	Software Testing Fundamentals & Documentation
Term Work 2	Test Case Design & Defect Reporting
Term Work 3	Manual Testing, Validation Techniques & Test Management
Term Work 4	Advanced Testing Techniques & AI-Assisted Software Testing
Term Work 5	Selenium Automation Testing with Python